```
    "Can the authors provide more details on the state and action space
used in Q-learning?",
    "How sensitive is the approach to the choice of hyperparameters for
Q-learning?",
    "Can the authors provide a more in-depth analysis of why Q-learning
leads to better performance?",
    "Can you provide more details on the implementation of the Q-learning
agent and its interaction with the training process?",
    "What specific benefits does Q-learning offer over other RL-based
hyperparameter optimization methods?",
    "Can you elaborate on the marginal improvements in validation loss? Why
are the differences so small?",
    "How does the proposed method generalize to other types of neural
network architectures or other hyperparameters?",
    "Can the authors provide more insights into the robustness and
generality of the proposed Q-learning based approach?",
    "How does the method perform on other types of neural network
architectures apart from transformers?",
    "Can the authors discuss potential limitations and ethical concerns in
more detail?"
],
"Limitations": [
    "The method's performance is sensitive to the choice of
hyperparameters, and there is additional overhead introduced by the
Q-learning agent.",
    "The experimental results do not convincingly demonstrate significant
improvements over baseline methods.",
    "The approach may not generalize well to other types of neural network
architectures without further tuning.",
    "The authors should discuss the potential drawbacks and challenges of
using Q-learning for learning rate adaptation in more detail.",
    "The paper does not adequately address the potential limitations and
ethical concerns of the proposed approach. It is important to discuss how
the method scales to other neural network architectures and the potential
risks associated with its use."
],
"Ethical Concerns": false,
"Soundness": 2,
"Presentation": 2,
"Contribution": 2,
"Overall": 3,
"Confidence": 4,
"Decision": "Reject"
```

D.7. Unlocking Grokking: A Comparative Study of Weight Initialization Strategies in Transformer Models

This idea was proposed in the 2nd iteration of a Sonnet 3.5 run.

Idea

```
"Name": "weight_initialization_grokking",
"Title": "Weight Initialization Grokking: Assessing the impact of weight
initialization strategies on the grokking phenomenon",
"Experiment": "Modify the `run` function to include different weight
initialization strategies (Xavier, He, orthogonal) for the Transformer
model. Specifically, adjust the model initialization phase in the
`Transformer` class to apply these strategies. Compare these against the
baseline (PyTorch default) by measuring the final training and validation
accuracy, loss, and the number of steps to reach 99% validation accuracy.
Evaluate the results for each dataset and seed combination.",
"Interestingness": 8,
"Feasibility": 7,
"Novelty": 7,
"novel": true
```

Link to code: https://github.com/SakanaAI/AI-Scientist/tree/main/example_papers/weight_initialization_grokking.

UNLOCKING GROKKING: A COMPARATIVE STUDY OF WEIGHT INITIALIZATION STRATEGIES IN TRANSFORMER MODELS

Anonymous authors
Paper under double-blind review

ABSTRACT

This paper investigates the impact of weight initialization strategies on the grokking phenomenon in Transformer models, addressing the challenge of understanding and optimizing neural network learning dynamics. Grokking, where models suddenly generalize after prolonged training, remains poorly understood, hindering the development of efficient training strategies. We systematically compare five initialization methods (PyTorch default, Xavier, He, Orthogonal, and Kaiming Normal) across four arithmetic tasks in finite fields, using a controlled experimental setup with a small Transformer architecture. Our approach combines rigorous empirical analysis with statistical validation to quantify the effects of initialization on grokking. Results reveal significant differences in convergence speed and generalization capabilities across initialization strategies. Xavier initialization consistently outperformed others, reducing steps to 99% validation accuracy by up to 63% compared to the baseline. Orthogonal initialization showed task-dependent performance, excelling in some operations while struggling in others. These findings provide insights into the mechanisms underlying grokking and offer practical guidelines for initialization in similar learning scenarios. Our work contributes to the broader understanding of deep learning optimization and paves the way for developing more efficient training strategies in complex learning tasks.

CAUTION: THIS PAPER WAS AUTONOMOUSLY GENERATED BY THE AI SCIENTIST

1 INTRODUCTION

Deep learning models have demonstrated remarkable capabilities across various domains, yet their learning dynamics often remain poorly understood Goodfellow et al. (2016). One intriguing phenomenon that has recently captured the attention of researchers is "grokking" Power et al. (2022). Grokking refers to a sudden improvement in generalization performance after prolonged training, often occurring long after the training loss has plateaued. This phenomenon challenges our understanding of how neural networks learn and generalize, particularly in the context of small, algorithmic datasets.

In this paper, we investigate the impact of weight initialization strategies on grokking in Transformer models Vaswani et al. (2017). While Transformers have become the de facto architecture for many natural language processing tasks, their behavior on arithmetic tasks provides a controlled environment to study fundamental learning dynamics. Understanding how different initialization methods affect grokking could provide valuable insights into optimizing model training and improving generalization performance.

Studying the relationship between weight initialization and grokking presents several challenges:

- Grokking itself is a complex phenomenon that is not fully understood, making it difficult to predict or control.
- The high-dimensional nature of neural network parameter spaces complicates the analysis of how initial weights influence learning trajectories.
- The interplay between initialization, model architecture, and task complexity adds another layer of intricacy to the problem.